

The Transport Layer: TCP

Lecture 12



UNIVERSITÀ DEGLI STUDI DI NAPOLI
FEDERICO II



Transport Layer

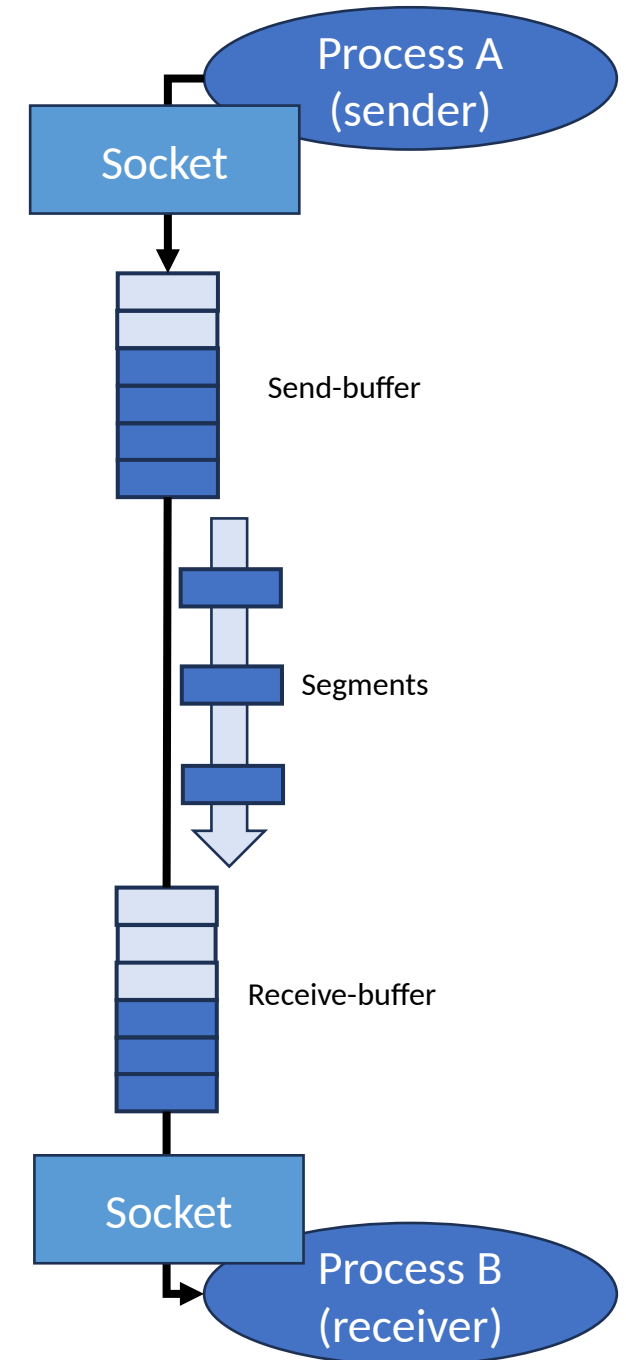
TCP

- TCP with respect to UDP is:
 - **Connection-oriented**: there is a “handshake” phase before the transmission that ensures the two processes (sender and receiver) to be “connected”.
 - **Reliable**: error detection, retransmissions, acknowledgments, timers, sequence numbers are implemented.
 - **Congestion and flow aware**: there is a regulation of the transmission rate depending on the receiver performance (flow) and the network performance (congestion).
- Connection orientation makes TCP full-duplex and point-to-point:
 - **Full-duplex**: if host A is connected with B, then B is connected with A.
 - **Point-to-point**: single sender and single receiver, i.e., the transfer of data from one sender to many receivers is not possible.
 - Note: UDP allows **multicasting**
- TCP is also oriented on sending/receiving **data streams**:
 - Multiple TCP segments may be part of a bigger data stream (ordered sequence of data).
 - Single UDP datagrams are considered mostly decoupled.

Transport Layer

TCP Buffers

- In TCP connection sending and receiving operations strongly **rely on buffers** to be performed.
- Buffers allow us to **partially decouple** transmission times from:
 - **Application delays**.
 - **OS delays** in multiplexing demultiplexing packets.
 - **Network delays** (oscillations of network performance).
- There is also the limit of the transmission rate, so long messages could **be broken into smaller segments** that may be collected in buffers before being disassembled (sender-side) or reassembled (sender-side).



Transport Layer

TCP Maximum Segment Size

- The amount of **data inside a segment** is limited by the **Maximum Segment Size (MSS)**:

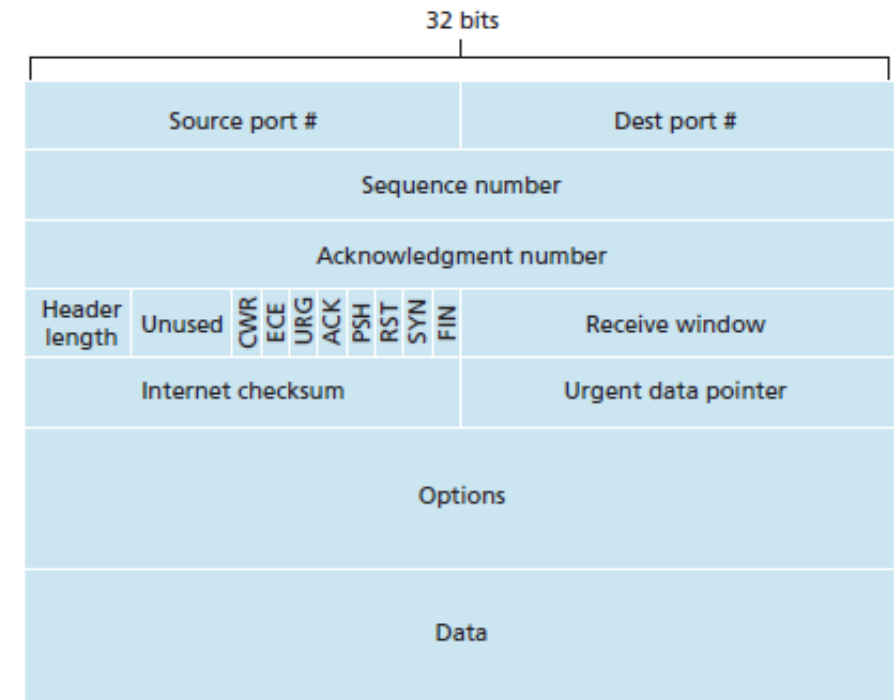
$$\text{MSS} + 40 = \text{MTU}$$

- Where:
 - The **Maximum Transmission Unit (MTU)** is the maximum length of a frame acceptable by the link-layer (e.g., in Ethernet it is 1500 bytes).
 - The **combined header size** of TCP and IP headers (typically 20+20 bytes).
- **On sending:** the segments are created from the application data and passed down to the network layer, where they are separately encapsulated within network-layer IP datagrams to be sent.
- **On receiving:** the data is placed into the receive-buffer, the application grabs data from the buffer when ready.

Transport Layer

TCP Segment

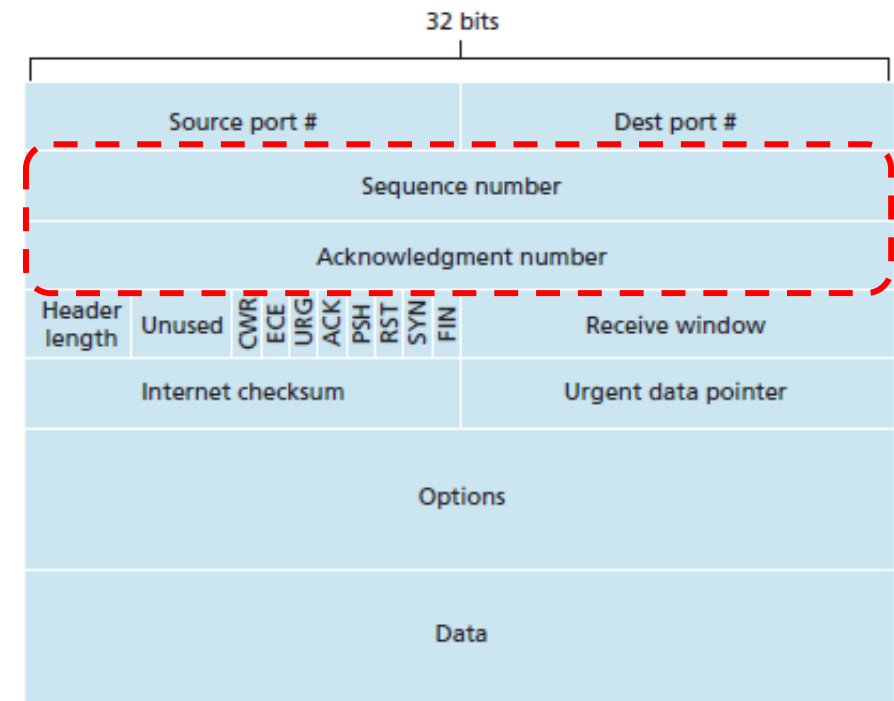
- The TCP segment consists of **header fields** and a **data field**, the latter contains some application data **of at most MSS size**.
- The **header** contains several fields:
 - **Sequence number** (32-bit) for reliable data transfer.
 - **Acknowledgment number** (32-bit) for reliable data transfer.
 - **Receive window** (16-bit) used to indicate the number of bytes that a receiver is willing to accept (for flow control).
 - **Header length** (4-bit) specifies the number of 32-bit words contained into the header. Note that **TCP header is variable due to the options field**.
 - **Options field** (K-bit) used for optional processes such as negotiating the MSS, time-stamping, etc.
 - **Flags** (6/12-bit) including:
 - ACK bit indicates that **this segment is an ACK** packet (so the acknowledgment number field is in use).
 - RST, SYN, and FIN bits used for **connection setup and teardown**.
 - CWR and ECE bits used in explicit congestion notification (optional).
 - PSH bit tells the receiver to pass the data to the application immediately. (RARE)
 - URG bit indicates that the segment contains “urgent” data. (RARE)
 - **Checksum** (16-bit): for integrity check.
 - **Urgent data pointer field** (16-bit) indicates the location of the last byte of the urgent part of the data. (RARE)



Transport Layer

TCP Sequence and Acknowledgment Numbers

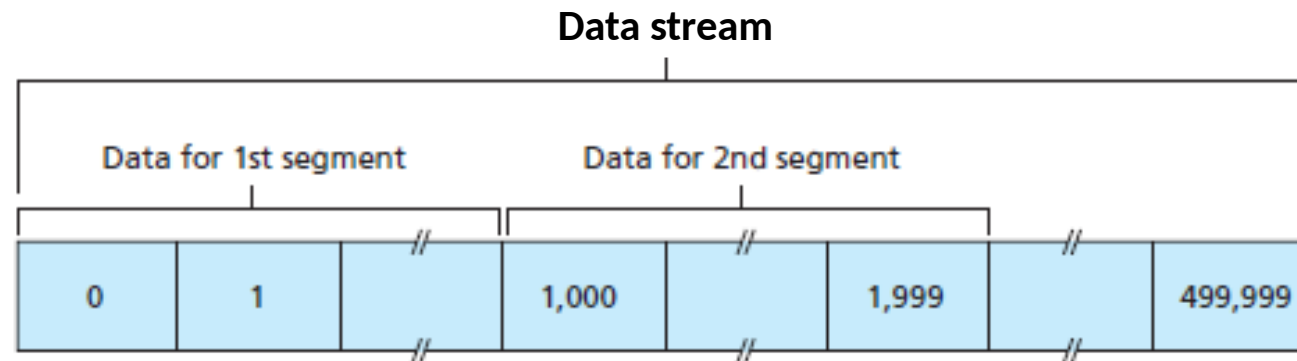
- Sequence numbers are used not only for reliable data transfer but **also to manage segmentation**.
 - If a large data stream is transmitted, we need **to break it into smaller “pieces”** depending on the MSS.
 - Each “piece” of the stream is put into a TCP segment.
- Sequence numbers and acknowledge numbers are **strictly related**.
 - The **sequence number** of a segment is conventionally the **position that the first byte of the segment’s data** has into the data stream.
 - The **acknowledgment number** is conventionally the **next piece of the data stream** we are expecting (calculated from the sequence number of the previous segment).



Transport Layer

TCP Sequence and Acknowledgment Numbers

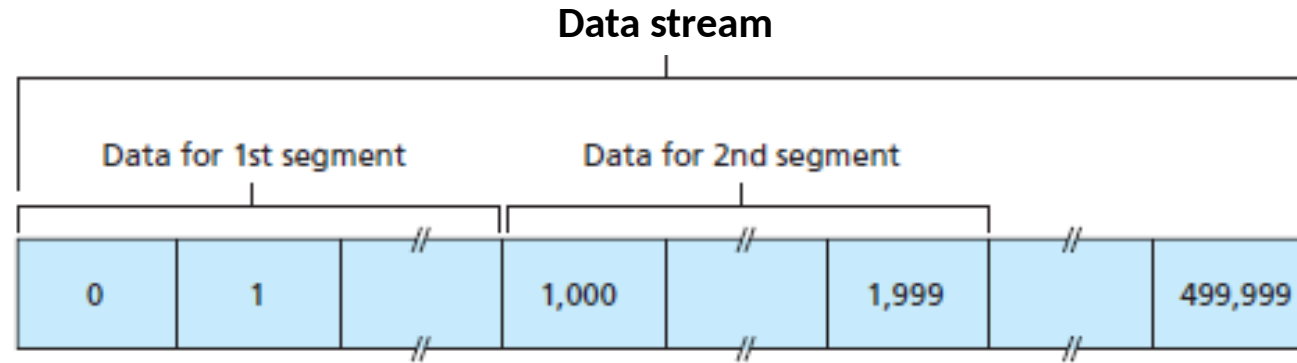
- Notice that TCP is a general-purpose transport protocol, it has no information about the data to be transmitted. From TCP's viewpoint data are **just an ordered stream of bytes**.
- The **sequence number** for a segment is then the byte-stream **number of the first byte of the data** in the segment.



- In the example, assuming a **data stream of 500,000 bytes** (~500KB), and a **MSS of 1,000 bytes**.
- The TCP constructs **500 segments** where the first segment has sequence number 0, the second segment has sequence number 1000, the third segment has sequence number 2000, and so on.

Transport Layer

TCP Sequence and Acknowledgment Numbers

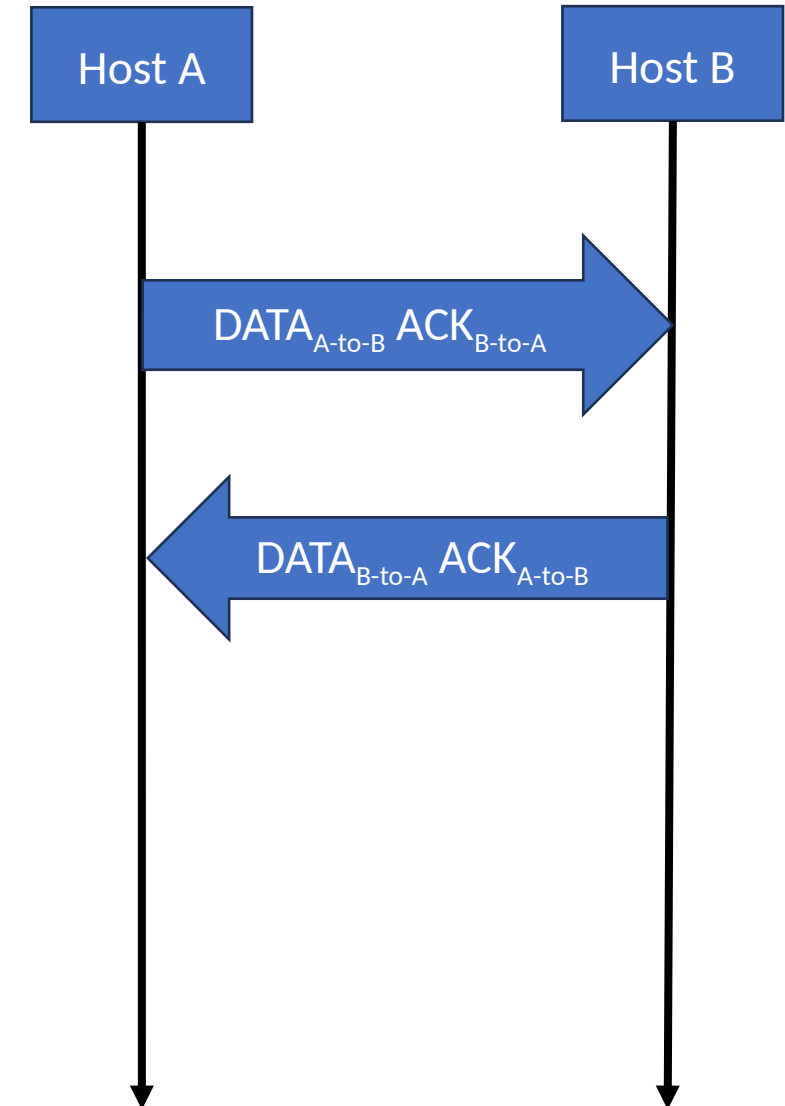


- Let's assume this data stream to be **passed from host A to host B**.
- For simplicity, let's **neglect previous interaction** (data exchange from connection establishment), so **we are starting from sequence number 0**:
 - The receiver will get a first segment of 1,000 bytes, i.e., from **sequence number 0 to sequence number 999**.
 - The receiver will then acknowledge the transmission by setting an **acknowledgment number of 1,000** (next expected byte in the stream).
 - The receiver will get the next segment of 1000 bytes, i.e., from **sequence number 1,000 to sequence number 1,999** and so on...

Transport Layer

TCP Sequence and Acknowledgment Numbers

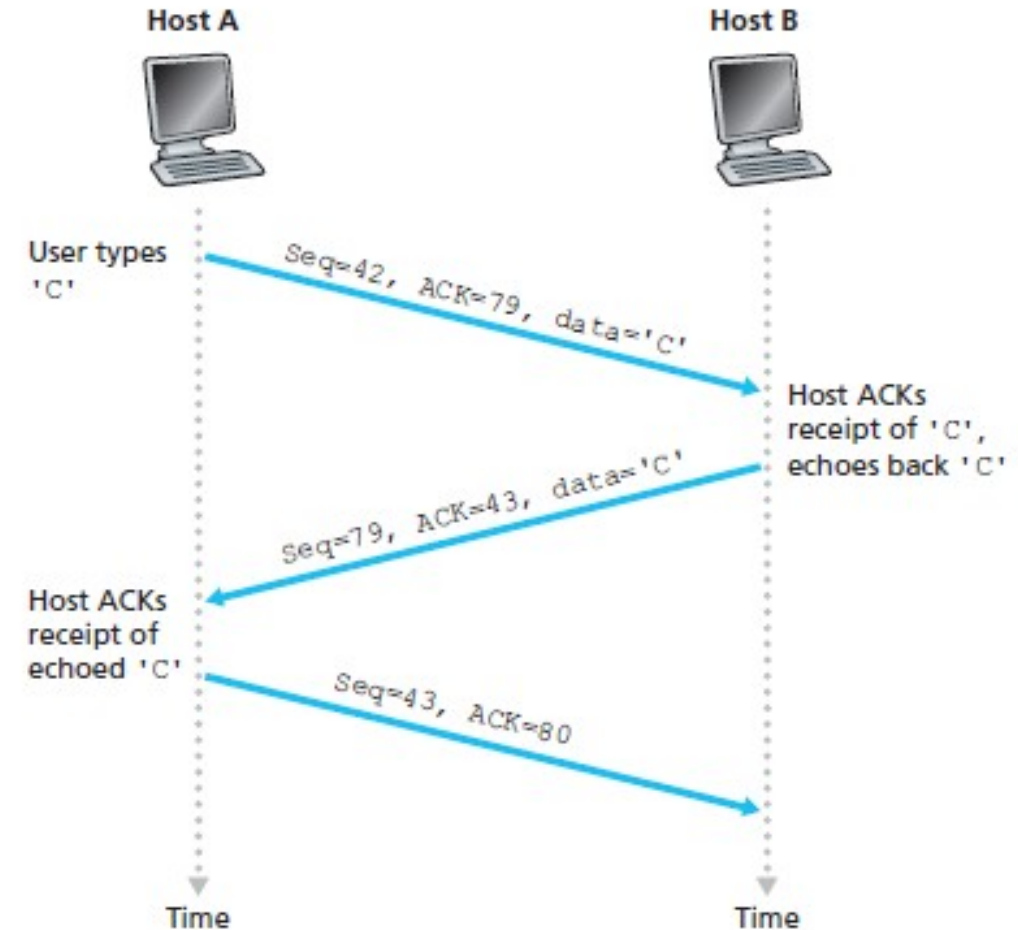
- What happens if the **receiver is sending data in return**?
 - Remind: TCP communication is **full-duplex**, if 2 hosts, A and B, communicate there are also 2 flows of data to be considered for reliable data transfer: **A-to-B** and **B-to-A**.
- In this case we can **use sequence numbers and acknowledgment numbers at the same time**:
 - Segments from A have **sequence** numbers related to **A-to-B data stream** and **acknowledge** numbers related to the **B-to-A data stream**.
 - Segments from B have **sequence** numbers related to **B-to-A data stream**, and **acknowledge** numbers related to the **A-to-B data stream**.



Transport Layer

TCP Sequence and Acknowledgment Numbers

- This is a simplified example of two hosts sending data each other.
 - We are considering 2 messages of **just 1 byte** (1 char) from A to B and vice versa.
 - Specifically, host A transmits a 'c' **that is echoed back** by B.
- Here that segments provide **ACK and DATA** **at the same time** (so the ACK flag is 1).



Transport Layer

TCP Sequence and Acknowledgment Numbers

1. From A to B:

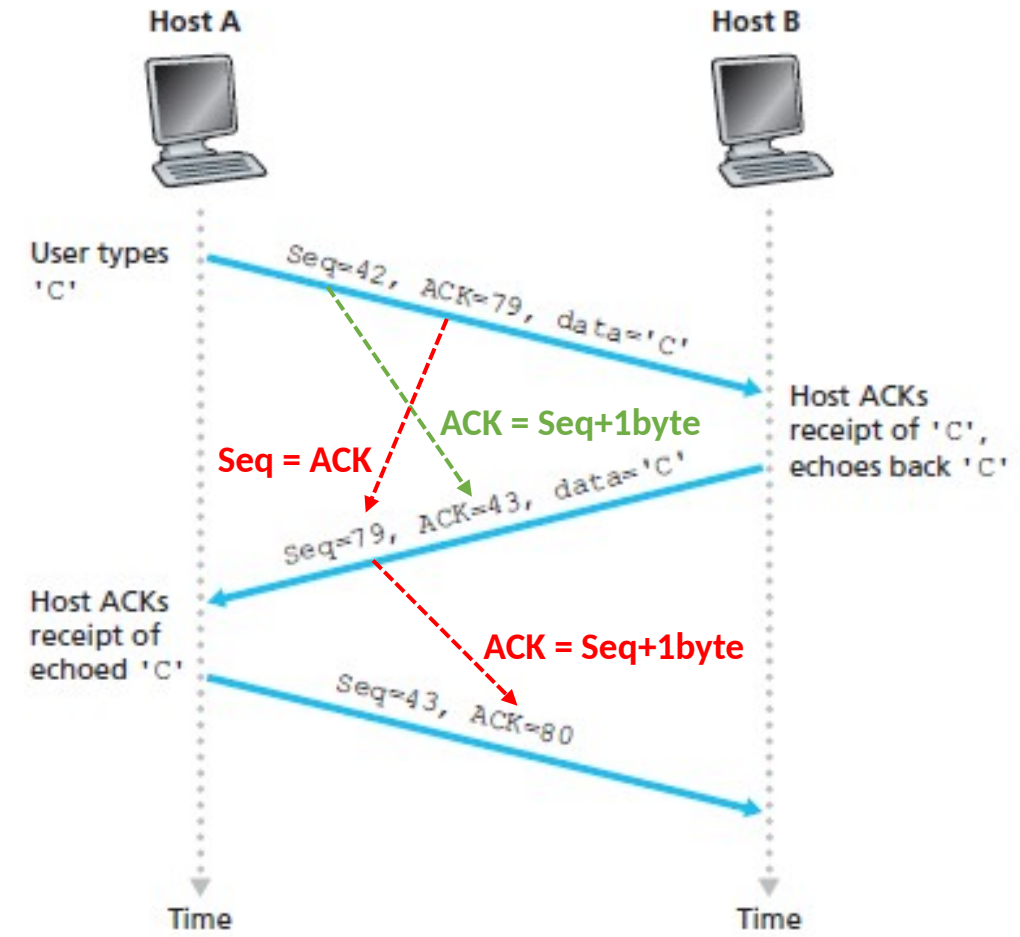
- The **ack number** as the **seq number of the next expected packet** in the B-to-A flow (byte 79).
- The **seq number** as the position of the **single byte** we are transmitting (byte 42).

2. From B to A:

- The **ack number** as the **seq number of the next expected packet** in the A-to-B flow (i.e., seq + 1 byte of the 'c' char).
- The **seq number** as the position of the **byte we are transmitting** (byte 79).

3. From A to B:

- Pure **ACK segment** is sent, having seq number 43 (as expected by B) and ack number 80 (next expected).



A-to-B info are in green.
B-to-A info are in red.

Transport Layer

Retransmission Timeout

- Previously we have emphasized that **TCP reliable data transfer mechanism makes extensive use of timeouts.**
- The default approach in pipelining-based TCP transmission is to **retransmit segments only if the associated ACKs are not received before the timeout.**
- Therefor, **timeout estimation is critical** as it strongly affects the performance of the transmission.
 - Too long timeout causes communication to be slowed.
 - Too short timeout causes the packets to overlap (and unnecessary packets to be retransmitted, producing useless traffic on the network).

Transport Layer

RTT Estimation

- To suitably set timeouts **we need a round-trip time (RTT) estimation.**
 - The timeout, i.e., the time to wait for a segment's ACK, should be **larger than RTT** otherwise unnecessary retransmissions would be sent.
- TCP estimates this value by **sampling the RTT of successfully acknowledged segments** that have not been retransmitted (one-shot).
- Since the time of a sampled RTT (SampleRTT) **may fluctuate** (due to traffic congestion, load of the receiver, etc.) the RTT estimation (EstimatedRTT) is given by the **exponential weighted moving average (EWMA)**:

$$\text{EstimatedRTT} = (1-\alpha) \cdot \text{EstimatedRTT} + \alpha \cdot \text{SampleRTT}$$

Where α is a parameter which is usually 0.125 (i.e., $1/8$).

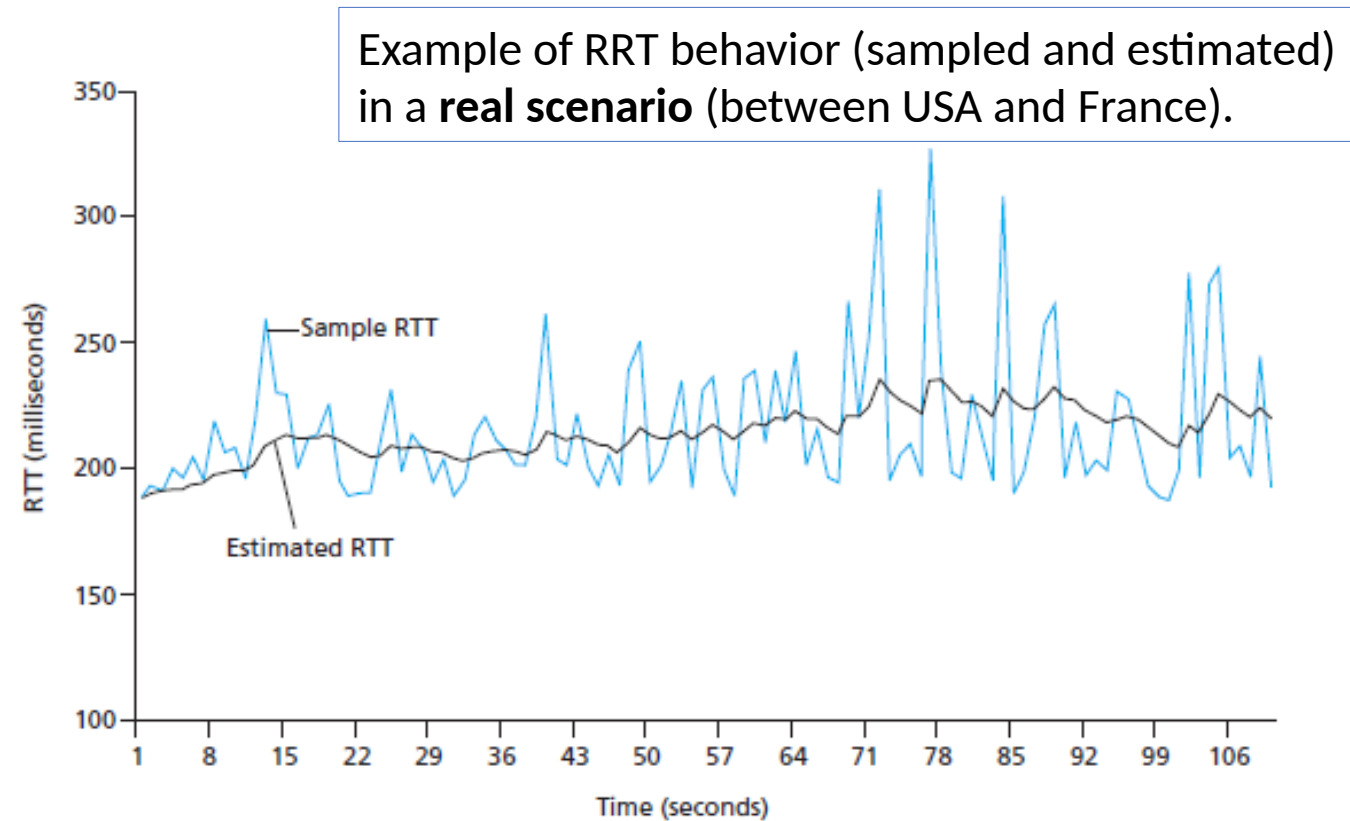
Transport Layer

RTT Estimation

- We can also consider the variability of the RRT in our timeout calculation.
- The **variability of the RTT** (DevRTT) can be given by the difference between the sample (SampleRTT) and the estimation (EstimatedRTT):

$$\text{DevRTT} = (1-\beta) \cdot \text{DevRTT} + \beta \cdot | \text{SampleRTT} - \text{EstimatedRTT} |$$

Where β is a parameter which is usually 0.25 (i.e., $1/4$).



Transport Layer

Retransmission Timeout

- Having these estimations, **it is possible to define a retransmission timeout** for TCP that is not too much lower or higher than the estimated RTT:

$$\text{TimeoutInterval} = \text{EstimatedRTT} + 4 \cdot \text{DevRTT}$$

- Here, the deviation of the RTT is used to set a **reasonable and adaptive margin** from the estimated RTT.
 - It increases when RTT oscillates, so the timeout window is larger **when we are uncertain** about our current RTT.
- By default, the **initial value** of the RTT (at $t=0$) is 1 second.

Transport Layer

Fast Retransmit

- Timeout-triggered retransmission is quite effective, but the **timeout period can be relatively long**.
- **Fast retransmit** is an approach in which the receiver signals the sender that one packet might be lost by sending **duplicate ACKs**.
 - When a TCP receiver receives a segment with a **sequence number that is larger than the expected one** it means that a segment has been reasonably missed.
 - The receiver **resends the old ACK** (duplicate ACK) having as ACK number the one of the expected segment.
 - If the **sender receives N duplicates** (typically 3 duplicates), it assumes that the previous segment has been lost so it is (fast) retransmitted **well before the deadline**.

Transport Layer

Fast Retransmit

- In this example we have a lost segment with sequence number 100.
 - Every time host B receives an out-of-order segment it **sends back a duplicated ACK** (ack=100).
 - When host A **receives 3 duplicates**, it assumes segment 100 to be lost and **resends it** (fast retransmit).
 - When host B finally receives the lost segment, **it will send the ACK of the next expected segment** depending on its own policy (go-back-N or selective repeat).
- But why don't we retransmit segments directly when the first duplicated ACK is received?
 - Because **we may receive wrong ACKs for different reasons** (not only if segments are lost).
 - For example, if **2 segments are received in the wrong order** (swapped during transmission), the **receiver will send 1 duplicate ACK** (for the first segment) before detecting the swap.

